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CHIPS: Efficient CLIP Adaptation via Curvature-aware Hybrid Influence-based Data Selection

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Abstract

*Adapting CLIP to vertical domains is typically approached by novel fine-tuning strategies or by continual pre-training (CPT) on large domain-specific datasets. Yet, data itself remains an underexplored factor in this process. We revisit this task from a data-centric perspective: Can effective data selection substitute for large-scale datasets in CPT? We introduce **CHIPS** (**C**urvature-aware **H**ybrid **I**nfluence in **P**rojection **S**ubspace), which assigns each image-text pair a utility score that integrates three complementary factors aligned with three goals: faithfulness via a curvature-aware and Newton-style alignment computed in CLIP's end-point subspace; scalability via an InfoNCE-aware curvature estimator with Johnson-Lindenstrauss (JL) sketching; and retention via a selection-aware relevance weight combined with learnability to balance target adaptation against general-domain preservation. We justify this design theoretically by proving a lower-bound guarantee on the proxy's correlation with full-parameter alignment and by characterizing the bias-variance trade-offs introduced by curvature mixing and JL sketching. We evaluate CHIPS empirically across various settings: 1) CHIPS attains state-of-the-art performance among selection baselines on **17 medical benchmarks**, matches full-dataset CPT with 30% of the data, and outperforms half-dataset CPT using only 10%; 2) on **31 general-domain benchmarks**, CHIPS yields the least performance drop under all retention ratios.*

1. Introduction

Vision-language models such as CLIP [44] achieve strong zero-shot recognition in general domains, but their performance degrades sharply in vertical settings (e.g., medical imaging and biology) where vocabulary, acquisition protocols, and label taxonomies shift substantially [66]. To adapt CLIP to such domains, two main paradigms have emerged: 1) *Model-centric* methods modify training or parameterization (e.g., probabilistic fine-tuning [25], many-to-many contrastive learning [24], and PEFT variants [18, 35, 46, 58, 65]); 2) *Data-centric* methods pursue continual pre-training (CPT) on large, domain-specific datasets, ranging from millions to hundreds of millions of pairs across medical and biodiversity settings [4, 15, 17, 23, 34, 36, 48, 50, 54, 59, 60, 64]. However, for vertical domains, collecting, curating, and processing increasingly large datasets is costly, and indiscriminate upsampling can even impair learning by introducing redundant, low-utility samples. This naturally raises a question: *Is sheer scale truly necessary for effective CPT?*

Meanwhile, a complementary line of work studies data attribution [9] to quantify how individual samples affect training dynamics and downstream generalization. Classical influence functions [28] and variants such as TracIn [43], EL2N [42], Arnoldi/GEX/FVM [27, 47, 62],

Method	Target	Sketch	Curvature	Domain	CLIP
Full CPT	✗	✗	✗	✗	✗
Random Selection	✗	✗	✗	✗	✗
Rule-filter	✗	✗	✗	✗	✗
CLIPscore	✗	✗	✗	✗	✗
Dot	✓	✓	✗	✗	✗
TracIn	✓	✓	✗	✗	✗
TRAK	✓	✓	✓	✗	✗
CHIPS (ours)	✓	✓	✓	✓	✓

Table 1. Properties of data selection methods for CLIP adaptation. *Target*: whether the strategy utilizes target evaluation set information. *Sketch*: whether JL Sketch is supported. *Curvature*: whether second-order information is employed. *Domain*: whether general-target domain balance is supported. *CLIP*: whether the method is specifically designed for CLIP models.

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were developed for single-tower model on supervised classification with additive cross-entropy and full-parameter updates. In CLIP, three mechanisms break these assumptions and systematically mis-rank examples. **(A) Cross-modal curvature of dual encoders.** CLIP's dual encoders induce non-block-diagonal second-order curvature and block-diagonal proxies ignore this coupling and mis-rank samples. **(B) Non-local gradients under InfoNCE.** Each sample's gradient depends on the softmax normalizer over the full negative set, making influence *batch-/global-dependent* rather than per-example additive and thus sensitive to batch size, queue design, and negative composition. **(C) End-point dominance at the projection heads.** Empirically, the projection heads and temperature parameter drive early shifts in similarity distributions, rendering full-parameter influence computation relatively unnecessary for CLIP.

Motivated by (A)-(C), we propose **CHIPS** (**C**urvature-aware **H**ybrid **I**nfluence in **P**rojection **S**ubspace), a CLIP-specific selector that scores each candidate by the expected one-step drop of the target evaluation loss and selects the top examples for CPT. CHIPS is designed for three goals, *faithfulness*, *scalability*, and *retention*, with one component per goal: (i) a Newton-style alignment computed on the projection heads and temperature that *provably lower-bounds* full-parameter alignment and enjoys better conditioning (Sec. 2.2); (ii) an InfoNCE-aware curvature preconditioner built from positive/negative gradient moments and compressed via Johnson-Lindenstrauss (JL) sketching to achieve near-linear time/memory with a quantified variance-bias trade-off (Sec. 2.3); and (iii) a selection-aware *soft* weighting that combines per-sample learnability with target-domain relevance to explicitly control the adaptation-retention balance (Sec. 2.4). Tab. 1 positions CHIPS against several common baselines from practical perspectives.

Across **17 medical benchmarks**, CHIPS *matches* full-dataset CPT using only 30% of the training pool, and *outperforms* half-dataset CPT using merely 10% data, achieving SOTA among selection baselines. For **31 general-domain benchmarks**, while domain adaptation inevitably reduces average performance, CHIPS consistently *reduces* the drop relative to the previous SOTA selector, indicating stronger retention of general-domain capabilities. In short, with principled attribution and CLIP-aware curvature, *effective CPT does not require extreme scale*.

Our contributions are threefold[1]. First, we propose an alignment score computed in end-point space of CHIP, which ranks samples by expected one-step decrease on evaluation loss. Second, we incorporate negative-pair geometry to capture cross-example curvature and analyze the variance-bias trade-off of the estimator. Third, we combine learnability and domain-relevance weights to emphasize decision-boundary, on-domain samples while softly limiting selection drift and mitigating catastrophic forgetting. Experimental results demonstrate the effectiveness of CHIPS on both medical and general downstream tasks.

[1] https://github.com/mihara-bot/CHIPS.

2. Method

CHIPS ranks training samples using an *alignment* score that estimates how much a one-step update on a given sample would reduce the evaluation loss on the target domain, together with auxiliary *learnability* and *domain relevance* scores. More specifically, CHIPS consists of three tightly coupled components: (i) a curvature-aware proxy alignment score computed in CLIP's end-point geometry, namely the projection heads and temperature; (ii) a negative-pair curvature estimator that captures the second-order coupling induced by symmetric InfoNCE and is made scalable through JL sketching; and (iii) two multiplicative weights for learnability and target-domain relevance, forming a single selection utility. Fig. 1 provides the workflow of CHIPS.

2.1. Problem Setup

We consider a pre-trained general CLIP model with parameters θ and a large target-domain training pool $\mathcal{D}_{\text{pool}} = \{z_i\}_{i=1}^{N}$ of image-text pairs. Our goal is to select a valuable subset from $\mathcal{D}_{\text{pool}}$ for CPT so as to maximize performance on a held-out target-domain evaluation set $\mathcal{D}_{\text{eval}}$ consisting of samples from various validation sets of downstream tasks. Let $\mathcal{L}_{\text{eval}}(\theta) \triangleq \mathbb{E}_{z\sim\mathcal{D}_{\text{eval}}}[\ell(z;\theta)]$ denote the evaluation loss, where ℓ is the symmetric InfoNCE objective [44] with a learnable logit-scale (temperature) $\tau > 0$. Denote the evaluation mean gradient by $\mathbf{u} \triangleq \nabla_\theta \mathcal{L}_{\text{eval}}(\theta) = \mathbb{E}_{z\sim\mathcal{D}_{\text{eval}}}[\nabla_\theta \ell(z;\theta)]$. During CPT, one Stochastic Gradient Descent (SGD) step draws a mini-batch of size B from a selection distribution q and computes the training gradient $\widehat{\mathbf{g}} = \frac{1}{B}\sum_{i=1}^{B} \nabla_\theta \ell(z_i;\theta)$ to update parameters via $\theta' = \theta - \eta\,\widehat{\mathbf{g}}$, where $\eta > 0$ is the learning rate. We also define the training mean gradient under q by $\mathbf{g}_q \triangleq \mathbb{E}_{z\sim q}[\nabla_\theta \ell(z;\theta)]$. Assume $\mathcal{L}_{\text{eval}}$ is twice continuously differentiable in a neighborhood of θ and let $\mathbf{H}_{\text{eval}}(\theta) = \nabla^2_\theta \mathcal{L}_{\text{eval}}(\theta)$. A second-order Taylor expansion with integral remainder and Hessian-Lipschitz constant L_H yields the following bound for $\Delta\theta = -\eta\,\widehat{\mathbf{g}}$:

$$\begin{aligned}\mathbb{E}\big[\Delta\mathcal{L}_{\text{eval}}\big] \;\le\; &-\eta\,\mathbf{g}_q^\top \mathbf{u} + \tfrac{1}{2}\,\eta^2\,\mathbb{E}\big[\,\widehat{\mathbf{g}}^\top \mathbf{H}_{\text{eval}}(\Theta)\,\widehat{\mathbf{g}}\,\big] \\ &+ \tfrac{L_H}{6}\,\eta^3\,\mathbb{E}\big[\|\widehat{\mathbf{g}}\|^3\big],\end{aligned} \tag{1}$$

for some Θ on the segment between θ and θ'. In a local quadratic model, the steepest descent direction is the Newton direction $\mathbf{H}_{\text{eval}}^{-1}\mathbf{u}$, and a good selection distribution q should **increase** the first-order alignment term $-\eta\,\mathbf{g}_q^\top\mathbf{u}$ while controlling curvature. For a more lightweight assumption, if $\mathcal{L}_{\text{eval}}$ is ρ-smooth (gradient-Lipschitz), then

$$\mathbb{E}[\Delta\mathcal{L}_{\text{eval}}] \;\le\; -\eta\,\mathbf{g}_q^\top\mathbf{u} + \tfrac{\rho}{2}\,\eta^2\,\mathbb{E}\big[\|\widehat{\mathbf{g}}\|^2\big]. \tag{2}$$

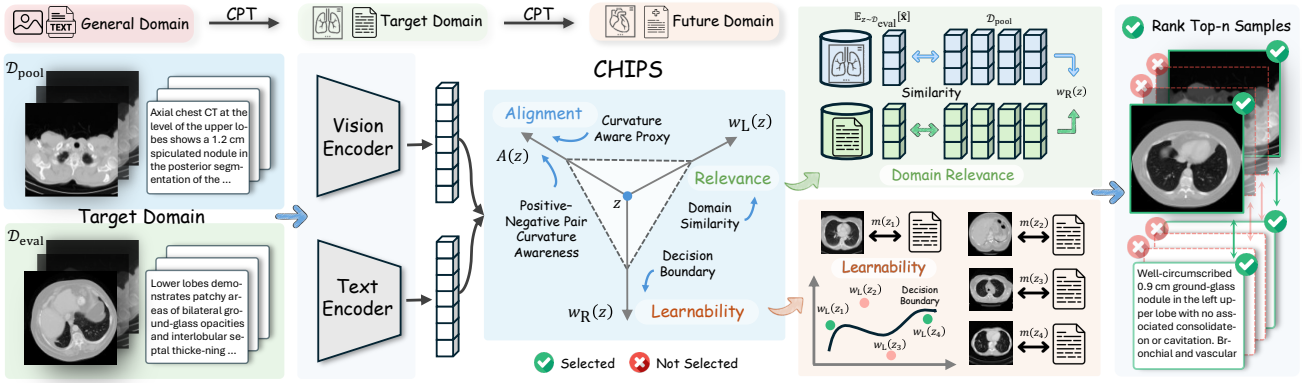


Figure 1. Workflow of CHIPS. For each training sample, CHIPS computes a curvature-aware proxy Newton alignment in CLIP's end-point space (projection heads and temperature), where curvature is approximated by mixing self and negative-pair cross moments from symmetric InfoNCE and scaled efficiently via JL sketching. The alignment is then modulated by learnability and target-domain relevance to yield a single selection utility, and the top-n samples are chosen to CPT CLIP models for domain adaptation.

Eq. (1) and Eq. (2) emphasize the same principle: *descent is driven by the alignment between the expected training gradient and the evaluation mean gradient*. While we present SGD for clarity, the alignment direction naturally extends to AdamW and full proofs are provided in App. B.1. These observations motivate us to prioritize samples whose updates align with (a proxy of) the Newton direction; the construction of such a curvature-aware proxy in CLIP's end-point geometry is given next in Sec. 2.2.

2.2. Curvature-aware Proxy Alignment

CHIPS is implemented on the end-point subspace of CLIP $\vartheta = \{\mathbf{W}_v, \mathbf{W}_t, \tau\}$, consisting of the two projection heads and the temperature. For a sample z, let $\mathbf{g}_\vartheta(z) = \nabla_\vartheta \ell(z;\theta)$ denote its gradient restricted to ϑ, and let $\mathbf{u}_\vartheta$ be the corresponding mean evaluation gradient. Motivated by the one-step descent view in Sec. 2.1, the ideal update direction on this subspace is the Newton direction $\mathbf{H}_\vartheta^{-1}\mathbf{u}_\vartheta$, where $\mathbf{H}_\vartheta$ denotes a population curvature on ϑ. Accordingly, we define the curvature-aware proxy alignment score:

$$A(z) \triangleq \mathbf{g}_\vartheta(z)^\top \mathbf{M}^{-1}\mathbf{u}_\vartheta, \tag{3}$$

where $\mathbf{M} \succ 0$ is a tractable curvature surrogate detailed in Sec. 2.3. A large positive $A(z)$ means that a one-step update on z moves the model more strongly along a descent direction for the evaluation loss.

Our central hypothesis is that alignment measured in the end-point subspace is a reliable proxy for alignment in the full parameter space. Eq. (3) can be rewritten as $A(z) = \tilde{\mathbf{g}}_\vartheta(z)^\top \tilde{\mathbf{u}}_\vartheta$ with $\tilde{\mathbf{g}}_\vartheta = \mathbf{M}^{-1/2}\mathbf{g}_\vartheta$ and $\tilde{\mathbf{u}}_\vartheta = \mathbf{M}^{-1/2}\mathbf{u}_\vartheta$. This indicates that the curvature-aware score is simply the standard inner-product alignment after a fixed, sample-independent preconditioning. Therefore, the proxy-full analysis below applies directly to this curvature-aware form. The explicit transformed covariance and coupling operators are given in App. B.2. To formalize this proxy view, consider the local linearization

$$\nabla_\theta \ell(z) = \mathbf{J}\,\mathbf{g}_\vartheta(z) + \mathbf{r}(z), \qquad \mathbf{u} = \bar{\mathbf{J}}\,\mathbf{u}_\vartheta + \boldsymbol{\varepsilon}, \tag{4}$$

where $\mathbf{J}$ is a local linear map that lifts a sample's end-point gradient $\mathbf{g}_\vartheta(z)$ to its full-parameter gradient, and $\bar{\mathbf{J}}$ is the analogous local linear map for the mean evaluation gradient. The residual terms $\mathbf{r}(z)$ and $\boldsymbol{\varepsilon}$ capture the corresponding approximation errors.

Define the subspace alignment $X(z) = \mathbf{g}_\vartheta(z)^\top \mathbf{u}_\vartheta$ and the full alignment $Y(z) = \nabla_\theta \ell(z)^\top \mathbf{u}$. Let $\Sigma_g = \mathrm{Cov}[\mathbf{g}_\vartheta(z)]$, $\mathbf{S} = \frac{1}{2}(\mathbf{J}^\top \bar{\mathbf{J}} + \bar{\mathbf{J}}^\top \mathbf{J})$, $\mathbf{B} = \Sigma_g^{1/2}\mathbf{S}\,\Sigma_g^{-1/2}$, and let $\sigma_\zeta^2 = \mathrm{Var}[\zeta(z)]$ denote the variance of the mismatch term induced by $(\mathbf{r}, \boldsymbol{\varepsilon})$. We have the following lower bound:

Theorem 1 (Proxy-full alignment correlation)**.** *If $\zeta(z)$ is uncorrelated with $\mathbf{g}_\vartheta(z)$, then the Pearson correlation between $X(z)$ and $Y(z)$ satisfies*

$$\rho_{XY} \geq \frac{\lambda_{\min}\big(\mathrm{sym}(\mathbf{B})\big)}{\sqrt{\|\mathbf{B}\|_2^2 + \sigma_\zeta^2 \,/\, \big(\mathbf{u}_\vartheta^\top \Sigma_g\, \mathbf{u}_\vartheta\big)}}. \tag{5}$$

Theorem 1 shows that the proxy score tracks the full score when two conditions hold: (i) the subspace-to-full map is well aligned along $\mathbf{u}_\vartheta$, as summarized by $\mathbf{S}$, and (ii) the residual mismatch variance σ_ζ^2 is not too large. Under approximate isotropy or commuting structure, the bound simplifies further (provided in App. B.2).

However, CHIPS uses the alignment score for *ranking* samples. Therefore, the relevant empirical quantity is the *Spearman correlation* between proxy and full scores. On a 100K subset of BIOMEDICA [36], we observe a Spearman correlation of 0.83, proving that end-point subspace scores preserve the ordering induced by full alignment well.

2.3. Curvature Estimation

Symmetric InfoNCE couples each positive pair with multiple negatives through the softmax normalizer, creating *cross-example* curvature that a positive-only or diagonal surrogate fails to capture. Since our alignment score $A(z)$ relies on a curvature-preconditioned inner product, missing this cross mass yields a biased Newton proxy and unstable rankings along highly coupled directions. Working in the end-point subspace ϑ, we approximate the population curvature by combining *self* and *cross* second moments of subspace gradients:

$$\begin{aligned}\Phi_{\text{pos}} &= \tfrac{1}{N}\sum_{i=1}^{N} \mathbf{g}_\vartheta(z_i)\,\mathbf{g}_\vartheta(z_i)^\top,\\ \Phi_{\text{neg}} &= \tfrac{1}{N(N-1)}\sum_{i\neq j} \mathbf{g}_\vartheta(z_i)\,\mathbf{g}_\vartheta(z_j)^\top,\end{aligned} \tag{6}$$

where Φ_{pos} captures *self* curvature, while Φ_{neg} restores the *cross* mass induced by random negatives in InfoNCE. We then form a trace- and spectrum-controlled surrogate

$$\mathbf{H}_\vartheta^{(\alpha)} = (1-\alpha)\,\Phi_{\text{pos}} + \alpha\,\Phi_{\text{neg}}, \qquad \mathbf{M} = \mathbf{H}_\vartheta^{(\alpha)} + \lambda\,\mathbf{I}, \tag{7}$$

with a mixing weight $\alpha \in [0,1]$ and a small Tikhonov parameter $\lambda > 0$ to ensure numerical stability in matrix inversion. Plugging $\mathbf{M}$ into Eq. (3) yields the *curvature-aware* proxy Newton score $A(z)$. By mixing Φ_{pos} with the cross moment Φ_{neg} (with $\alpha > 0$), we restore the off-diagonal curvature induced by InfoNCE's negatives that positive-only surrogates ignore. The surrogate $\mathbf{M} \succ 0$ thus preconditions $A(z)$ in a fixed metric without altering the proxy-full correlation framework (Sec. 2.2); α controls coupling strength and λ provides ridge stability. This yields an alignment score that is sensitive to the true second-order structure yet robust. Let the ideal scalar be $A^\star(z) = \mathbf{g}_\vartheta(z)^\top \mathbf{H}_\vartheta^{-1}\mathbf{u}_\vartheta$ and the curvature-approximated score $A_\alpha(z) = \mathbf{g}_\vartheta(z)^\top \mathbf{M}^{-1}\mathbf{u}_\vartheta$ with $\mathbf{M}$ as in Eq. (7), and define $\Delta_\alpha = \mathbf{H}_\vartheta - \mathbf{H}_\vartheta^{(\alpha)}$. For further efficiency, we apply a k-dimensional JL sketch to compute a sketched score $\widehat{A}_\alpha(z)$. We discuss the choice of JL sketch in Sec. 4.1.

Theorem 2 (Error bound for curvature mixing)**.** *Assume bounded spectra for $\mathbf{H}_\vartheta$ and $\mathbf{H}_\vartheta^{(\alpha)}$. Then, with probability at least $1-\delta$ over the JL sketch (vacuous if no sketch is used), there exist constants $C_1, C_2 > 0$, depending only on spectral bounds of $\mathbf{H}_\vartheta$ and $\mathbf{H}_\vartheta^{(\alpha)}$, such that*

$$\begin{aligned}\mathbb{E}_z\big[\big(\widehat{A}_\alpha(z) - A^\star(z)\big)^2\big] \le \underbrace{\frac{C_1\log(1/\delta)}{k}}_{\textit{projection variance}} \\ + \underbrace{C_2\,\|\Delta_\alpha\|_F^2\,\|\mathbf{H}_\vartheta^{-1}\mathbf{u}_\vartheta\|_2^2}_{\textit{curvature bias}}.\end{aligned} \tag{8}$$

Eq. (8) separates an $O(1/k)$ *projection variance* term and a *curvature bias* term that shrinks as $\mathbf{H}_\vartheta^{(\alpha)}$ better matches $\mathbf{H}_\vartheta$. Because Φ_{neg} restores off-diagonal mass whenever cross-example coupling is present, any $\alpha > 0$ that does so reduces $\|\Delta_\alpha\|_F$ and tightens the bound; λ stabilizes the inverse without changing the decomposition. All derivations and proofs are deferred to App. B.3.

2.4. Learnability and Domain Relevance

The curvature-aware alignment score $A(z)$ in Eq. (3) indicates whether a sample points in a *useful* descent direction, but it does not distinguish already-solved cases from borderline ones, nor does it compensate for the distributional gap between $\mathcal{D}_{\text{pool}}$ and $\mathcal{D}_{\text{eval}}$. Therefore, we introduce another two multiplicative weights to modulate $A(z)$ by *learnability* and *target-domain relevance*.

Learnability. For a mini-batch of paired image–text examples $\{(x_i, y_i)\}_{i=1}^B$, let the end-point embeddings be $\hat{\mathbf{x}}_i = \frac{\mathbf{W}_v\mathbf{h}_i}{\|\mathbf{W}_v\mathbf{h}_i\|}$ and $\hat{\mathbf{y}}_j = \frac{\mathbf{W}_t\mathbf{t}_j}{\|\mathbf{W}_t\mathbf{t}_j\|}$ with temperature $\tau > 0$. The CLIP similarity is $s_{ij} = \tau\,\hat{\mathbf{x}}_i^\top\hat{\mathbf{y}}_j$, with softmax probabilities $p_{ij}^{i2t} = \frac{\exp(s_{ij})}{\sum_{j'}\exp(s_{ij'})}$ and $p_{ij}^{t2i} = \frac{\exp(s_{ij})}{\sum_{i'}\exp(s_{i'j})}$. Define the average correctness and a hardest-negative margin as

$$\begin{aligned}p_{\text{corr}}(z) &= \tfrac{1}{2}\big(p_{ii}^{i2t} + p_{ii}^{t2i}\big),\\ m(z) &= s_{ii} - \max\Big\{\max_{j\neq i} s_{ij},\ \max_{i'\neq i} s_{i'i}\Big\}.\end{aligned} \tag{9}$$

We then weight toward decision-boundary examples and away from saturated ones:

$$w_{\text{L}}(z) = \big(1 - p_{\text{corr}}(z)\big)\,\big(1 + \sigma(-m(z))\big), \tag{10}$$

where $\sigma(\cdot)$ is the sigmoid function. The first factor down-weights already-correct and high-confidence pairs to reduce their impact while the second emphasizes small/negative margins that are most learnable in one step.

Target-domain Relevance. To softly favor samples consistent with $\mathcal{D}_{\text{eval}}$, we compute evaluation embeddings $\boldsymbol{\mu}_x = \mathbb{E}_{z\sim\mathcal{D}_{\text{eval}}}[\hat{\mathbf{x}}]$, $\boldsymbol{\mu}_y = \mathbb{E}_{z\sim\mathcal{D}_{\text{eval}}}[\hat{\mathbf{y}}]$, and define

$$w_{\text{R}}(z) = \sigma\Big((1-\beta)\,\cos(\hat{\mathbf{x}}, \boldsymbol{\mu}_x) + \beta\,\cos(\hat{\mathbf{y}}, \boldsymbol{\mu}_y)\Big), \tag{11}$$

where $\beta \in [0,1]$ controls the balance between two modalities and $\cos(\cdot,\cdot)$ denotes the cosine similarity. Notably, the relevance logit $s(z) = (1-\beta)\,\cos(\hat{\mathbf{x}}, \boldsymbol{\mu}_x) + \beta\,\cos(\hat{\mathbf{y}}, \boldsymbol{\mu}_y)$ in Eq. (11) lies in $[-1,1]$, therefore $w_{\text{R}}(z) \in [\sigma(-1), \sigma(1)] \approx [0.27, 0.73]$. This implements soft re-weighting to some extent rather than hard filtering as no sample is zeroed out.

Hence, w_{R} acts as a gentle target-domain bias rather than a hard filter: it nudges selection toward target-relevant samples while preventing the final distribution from drifting too

far from the original alignment-driven one. This helps balance adaptation and retention, mitigating catastrophic forgetting of general-domain knowledge.

Final utility score and selection. We combine the components multiplicatively into a single ranking score.

$$\mathcal{I}_{\text{CHIPS}}(z) = \widehat{A}_{\alpha}(z) \cdot w_{\text{L}}(z) \cdot w_{\text{R}}(z), \tag{12}$$

and select the top-n examples from $\mathcal{D}_{\text{pool}}$ for CPT. Implementation details of CHIPS are provided in App. C.

3. Experiment

3.1. Experimental Setup

Training. For a comprehensive comparison, we perform continual pre-training using the initialization weights of MetaCLIP-B32/B16/L14/H14 [57]. We select **medical** as target domain and two recently introduced large-scale medical multimodal datasets are employed as training pools: BIOMEDICA (24M samples) [36] and MedTrinity (18M samples) [56]. We select data retention ratio r as 10%, 20%, 30%, and 50% of the complete dataset, with the number of training epochs fixed at 5. All models are optimized using the AdamW optimizer ($\beta_1 = 0.9, \beta_2 = 0.98, \epsilon = 10^{-6}$) and a cosine learning rate scheduler (initial learning rate as 10^{-6}), with a global batch size of 32,768. Each experiment is conducted on an Ubuntu 24.04 server equipped with 8x NVIDIA H200 (141GB) GPUs. Further details regarding the training datasets, model architectures, and specific hyperparameter settings are provided in App. E.

Evaluation. We evaluate 48 tasks spanning general and medical domains to enable a comprehensive assessment. The general-domain suite includes Cars [29], Country211 [44], Fer2013 [16], Aircraft [37], Food101 [3], GTSRB [49], ImageNet-A, ImageNet-O [20], ImageNet-1K [8], ImageNetV2 [45], MNIST [31], Rendered-SST2 [44], STL-10 [7], SUN397 [55], VOC [12], Caltech-101 [13], CIFAR-10, CIFAR-100 [30], CLEVR (Closest-Object-Distance/Count-All) [26], DTD [6], EuroSAT [19], Oxford Flowers [39], KITTI [14], Pets [41], RESISC45 [5], smallNORB (Azimuth), smallNORB (Elevation) [32], SVHN [38], Flickr8k [22], Flickr30k [63], and MSCOCO [33]. For the medical domain, we consider 17 classification tasks across eight specialties. These include Ophthalmology (Diabetic [11], OCTMNIST, RetinaMNIST [61]); Radiology (ChestMNIST, ChestX-ray14 [52], OrganAMNIST, OrganCMNIST, OrganSMNIST); Dermatology (DermaMNIST); Hematology (BloodMNIST); Pathology (PCAM [51], LC25000 [2], PathMNIST); Neuropathology (Amyloid CAA/Diffuse [53]); Histology (TissueMNIST); and Non-clinical Biology (Pollen [1]). For classification tasks, the average accuracy is reported. For retrieval tasks, bidirectional (image-to-text and text-to-image) performance is evaluated using $R@1$ recall metric. Further details are provided in App. F.

Baselines. We select a total of seven data selection baselines for comparison, including (1) **random selection**, which selects samples uniformly from the training pool without any selective control; (2) three **heuristics-based methods**: CLIPScore [21], Concept-Balance, and Concept-Filter [36]; (3) **Influence-based methods**: Dot [67], TracIn [43], and TRAK [40]. Moreover, we also include three commonly used medical-specific CLIP models for reference: PubMedCLIP [10], BioMedCLIP [64], and BMCLIP [36]. Implementation details of all baseline methods and our CHIPS method are provided in App. C.

3.2. Main Results

Effectiveness of CHIPS. As shown in Tab. 2, across all retention ratios, CHIPS achieves the best average performance on 17 medical tasks among data selection methods, with Medical Avg scores of 27.03, 28.20, and 29.96, exceeding the second-best method by +0.57, +1.57, and +3.68 points, respectively. Notably, with only 10% of the full BIOMEDICA, CHIPS outperforms a 50% random subset (27.03 vs. 26.26); with 30%, it achieves **95.1%** of the full-dataset performance (29.96 vs. 31.51). It is also competitive with specialized medical CLIP models pre-trained on larger datasets: at $r = 30\%$, CHIPS slightly surpasses BMCLIP (29.96 vs. 29.86) and consistently outperforms PubMedCLIP and BioMedCLIP across r. Even though CHIPS is not uniformly best on every specialty, it yields the highest overall average. Importantly, CHIPS better preserves general-domain abilities than the previous SOTA selection method TracIn: for *CLS*, CHIPS retains 90.1%, 89.3%, and 87.2% of the vanilla model at $r = 10\%, 20\%, 30\%$ (vs. 88.9%, 88.0%, 87.0% for TracIn), and for *RET* it is comparable, slightly lower at $r = 10\%$ but higher at $r \geq 20\%$, yielding a higher average RET across retention ratios. Overall, CHIPS delivers the strongest medical performance while incurring *less* degradation on the general domain than TracIn.

Generalization abilities of CHIPS. With a single scoring pass on MetaCLIP-B16-400M (10% kept), we reuse the computed CHIPS scores to train diverse backbones (B32/B16/L14/H14) and pre-training scales (*400M* and *CC*). As shown in Tab. 3, across all seven settings, CHIPS attains the best Medical performance, outperforming TracIn by 0.20-2.65 points. On general-domain metrics, CHIPS typically ranks second (behind Random) and generally matches or exceeds TracIn on CLS and RET, with a single RET exception on B16-400M. Thus, CHIPS transfers across architectures and scales, allowing cached scores to

Model	Medical									General	
	OPH	RAD	DER	HEM	PAT	NEU	HIS	BIO	Avg	CLS	RET
PubMedCLIP	35.92	20.27	9.63	8.45	25.77	60.50	8.45	25.27	26.21	19.05	24.21
BioMedCLIP	14.42	15.59	1.20	10.14	21.17	44.79	4.66	72.91	21.45	7.68	23.55
BMCLIP	38.47	17.44	61.65	9.44	49.75	38.26	5.02	17.28	29.86	57.97	28.27
Vanilla	15.91	19.76	10.82	17.95	30.09	57.96	5.24	10.56	23.51	53.15	29.23
Full Dataset	33.30	12.75	11.72	23.33	49.57	82.79	12.40	9.81	31.51	49.72	24.20
r=10%											
Random	36.01	9.93	11.27	20.61	39.71	43.96	4.68	10.18	24.78	**52.21**	29.28
Concept-Balance	33.18	12.11	11.67	21.81	40.40	46.82	5.03	10.87	25.50	51.95	30.14
Concept-Filter	31.40	10.81	11.72	23.41	43.14	44.53	5.14	11.06	25.15	51.81	29.19
CLIPScore	15.48	12.75	11.47	20.99	38.23	64.15	5.25	10.56	24.16	53.39	**31.27**
Dot	42.01	8.54	12.17	16.87	36.98	45.49	4.16	23.51	25.32	48.27	26.47
TracIn	41.30	10.47	12.02	16.69	35.69	51.79	4.96	26.34	26.46	47.26	26.36
TRAK	30.29	12.37	12.17	25.93	39.04	45.95	4.46	12.88	25.19	48.24	27.17
CHIPS (ours)	38.93	11.05	11.87	21.43	37.99	50.22	5.29	25.96	**27.03**	47.88	25.71
r=20%											
Random	35.79	13.82	11.62	21.92	39.35	36.41	6.60	11.13	25.00	**51.39**	29.57
Concept-Balance	32.77	13.62	11.97	24.00	44.24	43.16	5.42	10.37	26.20	51.38	**30.31**
Concept-Filter	31.36	11.70	11.97	25.46	44.98	35.85	6.42	15.21	25.23	51.25	29.54
CLIPScore	17.33	9.83	11.27	18.82	38.01	32.35	4.45	11.25	20.01	51.25	28.81
Dot	39.20	11.19	11.77	13.30	35.46	53.72	5.38	33.63	26.39	47.50	26.93
TracIn	42.47	12.28	11.82	19.70	37.92	46.13	5.07	19.17	26.63	46.79	25.09
TRAK	43.25	10.12	11.02	16.75	37.66	28.83	5.41	28.16	24.54	47.54	25.81
CHIPS (ours)	44.13	13.02	11.87	22.49	40.79	38.54	5.92	30.36	**28.20**	47.45	25.90
r=30%											
Random	36.77	11.90	12.07	18.42	43.35	47.08	6.42	10.94	26.28	50.98	**29.85**
Concept-Balance	31.95	11.88	12.07	24.61	42.52	35.88	6.48	10.43	24.56	**51.13**	29.66
Concept-Filter	32.37	11.23	12.02	25.96	41.40	41.02	6.81	21.06	25.37	50.63	29.54
CLIPScore	21.93	8.25	11.32	13.94	38.81	26.49	4.49	11.94	19.01	50.76	28.09
Dot	38.42	11.05	11.62	15.40	37.52	35.04	7.69	20.74	23.97	46.39	25.67
TracIn	39.63	14.18	11.97	14.24	38.91	41.09	5.69	22.19	25.68	46.23	25.15
TRAK	38.95	10.49	11.42	16.92	36.96	25.70	7.75	25.14	23.54	46.29	24.04
CHIPS (ours)	41.66	13.56	11.37	25.02	43.10	46.53	7.69	36.34	**29.96**	46.34	26.11
r=50%											
Random	34.47	13.86	11.82	18.82	45.82	41.00	10.06	10.56	26.26	50.56	29.29

Table 2. Downstream performance of MetaCLIP-B16-400M [57] and variants continually pre-trained with data subsets from BIOMEDICA [36] selected by different methods, under a fixed retention ratio $r \in \{10\%, 20\%, 30\%, 50\%\}$. Abbreviations: for medical specialties, OPH = Ophthalmology, RAD = Radiology, DER = Dermatology, HEM = Hematology, PAT = Pathology, NEU = Neuropathology, HIS = Histology, BIO = Non-clinical Biology; for metrics in general domain, *CLS* denotes classification accuracy, and *RET* denotes the average R@1 across image-to-text and text-to-image retrieval tasks. In each setting, the best result is **bolded** and the second best is underlined.

be reused while delivering strong medical gains with minimal loss of general ability.

Ablation study. As shown in Tab. 4, we ablate CHIPS by progressively adding components from Eq. (12). On the medical benchmarks, CHIPS achieves the highest performance at all budgets, surpassing the strongest ablation by +1.05, +0.28, and +1.46 points at $r = 10\%, 20\%, 30\%$, respectively. These gains support the multiplicative combination of alignment, learnability, and domain relevance, with the relevance term becoming especially beneficial at larger budgets ($r = 30\%$). Meanwhile, general-domain performance remains close to the best ablation (within $\leq$0.53 for General CLS and $\leq$0.99 for General RET), indicating minimal trade-off for the medical improvements. On the general domain, the retrieval gap to the best ablation narrows as r grows (0.99 $\rightarrow$ 0.66 $\rightarrow$ 0.37 for General RET), suggesting controlled specialization towards the target domain rather

Model	Medical CLS	General CLS	General RET
B32-400M			
Random	27.15	**49.31**	**27.33**
TracIn	27.48	47.19	25.10
CHIPS (ours)	**27.83**	47.90	25.65
B32-CC			
Random	27.95	**50.69**	**29.28**
TracIn	26.25	48.90	25.72
CHIPS (ours)	**28.13**	49.48	26.79
B16-400M			
Random	24.78	**52.21**	**29.28**
TracIn	26.46	47.26	26.36
CHIPS (ours)	**27.03**	47.88	25.71
B16-CC			
Random	26.30	**55.91**	**30.16**
TracIn	25.89	50.17	27.37
CHIPS (ours)	**26.93**	51.27	28.17
L14-400M			
Random	29.33	**57.07**	**33.35**
TracIn	27.08	52.99	28.17
CHIPS (ours)	**29.73**	53.65	28.17
L14-CC			
Random	30.75	**60.74**	**32.34**
TracIn	31.54	56.98	28.26
CHIPS (ours)	**31.74**	57.93	30.05
H14-CC			
Random	35.23	**61.36**	**32.82**
TracIn	35.10	58.27	31.60
CHIPS (ours)	**35.48**	58.24	32.09

Table 3. Generalization experiment of MetaCLIP-series adapted models continually pre-trained on 10% kept data from different data selection methods. *400M* denotes the model was pre-trained on 400M general image-text pairs and *CC* denotes the model was pre-trained on 2.5B general image-text pairs.

than catastrophic forgetting.

Cost analysis. We measure scoring cost as total FLOPs over $\mathcal{D}_{\text{pool}}$. CHIPS requires 50.9475×10^{15} FLOPs, 3.1% lower than TracIn (52.5891×10^{15}) and effectively on par with TRAK (50.9458×10^{15}). Despite equal-or-lower cost, CHIPS yields consistently stronger medical performance than TracIn (Tab. 2), improving by +0.57, +1.57, and +4.28 points across three retention settings, respectively. In practice, the one-time scoring overhead is further amortized because CHIPS scores can be cached and reused across architectures and pre-training scales (Tab. 3). The full FLOPs computation process is provided in App. G.

4. Analysis

4.1. Alignment

Effect of target evaluation set size. Expanding the target evaluation set $\mathcal{D}_{\text{eval}}$ sharpens the alignment direction $\mathbf{u}_\vartheta$ and improves medical accuracy with minimal general-domain drift. As shown in Fig. 2(A), as samples per task increase from 50 to 200, Medical CLS improves from 26.25 to 27.03 (+0.78), then saturates at 250 (27.12, +0.09). The slight dip at 100 (25.82) reflects higher variance at smaller set sizes, while overall the trend is upward and stabilizing around 200. General CLS remains essentially flat across all sizes (48.41 at 50 to 47.88 at 200; total range $\leq$0.55), indicating that larger $\mathcal{D}_{\text{eval}}$ chiefly benefits medical alignment without harming general performance. Balancing diminishing returns beyond 200 against the linear increase in scoring cost, and we adopt 200 samples per task as the default.

Model	Medical CLS	General CLS	General RET
r=10%			
Alignment-only	25.98	48.33	**26.70**
Alignment-Margin	25.95	**48.41**	26.25
CHIPS (ours)	**27.03**	47.88	25.71
r=20%			
Alignment-only	27.52	46.67	26.25
Alignment-Margin	27.92	**46.88**	**26.56**
CHIPS (ours)	**28.20**	46.73	25.90
r=30%			
Alignment-only	27.84	**46.78**	25.23
Alignment-Margin	28.50	46.75	**25.52**
CHIPS (ours)	**29.96**	46.34	25.15

Table 4. Ablation experiment of CHIPS on MetaCLIP-B16-400M under 10%, 20%, and 30% data retention ratios. *Alignment-only* uses single alignment score, *Alignment-Margin* additionally introduces the margin term, and CHIPS multiplies all three metrics.

Projection layers dominate alignment with text being more informative. We ablate the end-point geometry $\vartheta = \{\mathbf{W}_v, \mathbf{W}_t, \tau\}$ into *All*, *Logit-only*, *Visual-only*, and *Text-only*, with results shown in Fig. 3. Across methods, *Text-only* nearly matches *All* (98.6–99.7%), *Visual-only* lags, and *Logit-only* is weakest. For CHIPS, the scores are 27.03 (*All*), 26.95 (*Text-only*, -0.08; 99.7%), 26.67 (*Visual-only*, -0.36; 98.7%), and 23.83 (*Logit-only*, -3.20). These results show that alignment resides primarily in the projection heads, especially $\mathbf{W}_t$, with $\mathbf{W}_v$ providing a smaller yet complementary gain. Moreover, CHIPS is more robust to projection layers compared with baseline methods, retaining 99.7% (*Text-only*) and 98.7% (*Visual-only*), the highest among methods, and exhibiting the smallest text–visual gap (0.28 vs. 2.43/3.77/1.05 for Dot/TracIn/TRAK).

Effect of random projection. We compare three random projection methods in computing the alignment score $\widehat{A}_\alpha(z)$ and vary the target dimension $k \in \{2048, 4096, 8192, 16384\}$, with results shown in Fig. 4. *Sparse* projections are the most effective, peaking at *28.31* (16k) and remaining strong even at 2k (26.56). *CountSketch* improves with dimension (24.40→27.64 from 2k to 16k), with a small dip at 8k (26.12); at 4k it already reaches

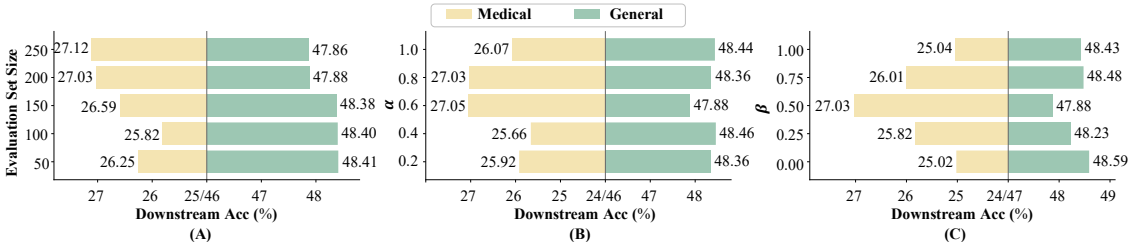


Figure 2. Downstream results of MetaCLIP-B16-400M continually pre-trained on 10% data selected from CHIPS under: **(A)** different evaluation set sizes, **(B)** different mixing α in computing alignment scores, and **(C)** different balance β in computing relevance scores.

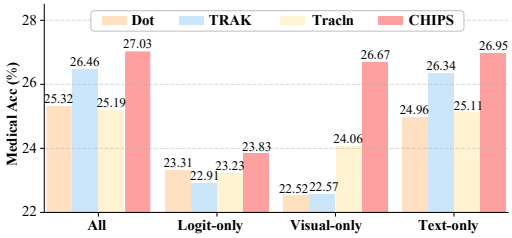


Figure 3. Medical downstream results of MetaCLIP-B16-400M continually pre-trained on 10% data selected from different selection methods under various end-point geometry settings.

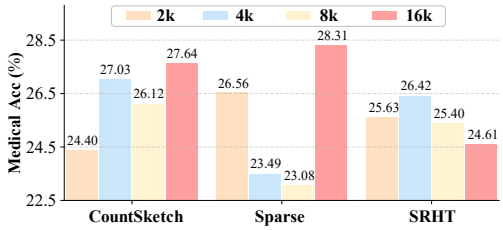


Figure 4. Medical downstream results of MetaCLIP-B16-400M continually pre-trained on 10% data selected from CHIPS under various random projection settings.

27.03. *SRHT* is less robust, peaking at 4k (26.42) and degrading at 16k (24.61). Overall, *Sparse*-16k is best (+0.67 over *CountSketch*-16k; +1.89 over *SRHT*-4k). For tighter budgets, *CountSketch*-4k offers a cost-efficient alternative within about 1.3 points of the best.

Balancing positive and negative curvature. In Eq. (7), α linearly mixes positive and negative pair curvature to form $\mathbf{M}$, which reweights gradients in $A(z) = \mathbf{g}_\vartheta(z)^\top \mathbf{M}^{-1}\mathbf{u}_\vartheta$. As shown in Fig. 2(B), Medical CLS peaks at α=0.6 (27.05) and remains essentially tied at 0.8 (27.03), while both smaller (0.2: 25.92; 0.4: 25.66) and larger (1.0: 26.07) values underperform by about 1-1.4 points. General CLS is almost flat across α (48.36-48.46), with a modest dip at 0.6 (47.88), indicating that α mainly governs target-domain discrimination with negligible general-domain impact. Mechanistically, moderate α balances attraction to positives and repulsion from negatives, improving the conditioning of $\mathbf{M}$ and yielding a more discriminative alignment direction; extremes overweight one term and reduce $A(z)$'s utility. We therefore recommend $\alpha \in [0.6, 0.8]$ (default 0.6 for medical focus; 0.4 if slightly prioritizing general-domain retention).

4.2. Domain Relevance and General Retention

In Eq. (11), β balances visual and textual similarities when computing $w_{\mathrm{R}}(z)$. As shown in Fig. 2(C), varying β reveals a clear adaptation-retention trade-off: Medical CLS peaks at β=0.5 (27.03), improving by 2.01-1.21 points over the unimodal extremes (β=0: 25.02; β=1: 25.04). In contrast, General CLS is flattish (48.59→47.88 across settings; total spread 0.71), with the best retention at the extremes (β=0: 48.59; β=1: 48.43) and the largest dip at β=0.5 (47.88). This U-shaped retention pattern indicates that balanced relevance sharpens specialization to the medical target, whereas unimodal weighting selects more generic samples that better preserve general-domain ability. A slight skew toward text (β=0.75) offers a good compromis: 26.01 on Medical (within 1.02 of the best) with near-top 48.48 on General, which is consistent with our end-point geometry ablation where text signals are more informative (Sec. 4.1). Finally, the overall narrow range in General CLS ($\leq$0.71) aligns with the bounded-drift property of w_{R}, which limits deviation from the base alignment distribution. We adopt β=0.5 for maximal target-domain gains.

5. Conclusion

We propose CHIPS, showing that principled data selection can substitute for raw scale in CLIP continual pre-training. On 17 medical benchmarks, CHIPS attains SOTA performance among selection baselines, matches full-dataset CPT with 30% of the data, and outperforms half-dataset CPT using only 10%. Across 31 general-domain benchmarks, it yields the smallest performance drop under 10–30% retention ratios, better preserving general capabilities. Limitations include reliance on a target validation distribution, we will explore unlabeled or shift-robust target signals and extensions beyond the CLIP model in the future work.

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